

```
1 from google.colab import files
2 uploaded = files.upload()
```

Choose Files

WA_Fn-Us...er-Churn.csv

WA_Fn-UseC_-Telco-Customer-Churn.csv(text/csv) - 977501 bytes, last modified: 9/27/2019 - 100% done

Saving WA_Fn-UseC_-Telco-Customer-Churn.csv to WA_Fn-UseC_-Telco-Customer-Churn.csv

LOADING DATASET

```
1 import pandas as pd
2
3 df = pd.read_csv('WA_Fn-UseC_-Telco-Customer-Churn.csv')
4
5 df.head()
```

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecurity
0	7590-VHVEG	Female	0	Yes	No	1	No	No phone service	DSL	No
1	5575-GNVDE	Male	0	No	No	34	Yes	No	DSL	Yes
2	3668-QPYBK	Male	0	No	No	2	Yes	No	DSL	Yes
3	7795-CFOCW	Male	0	No	No	45	No	No phone service	DSL	Yes
4	9237-HQITU	Female	0	No	No	2	Yes	No	Fiber optic	No

5 rows × 21 columns

EXPLORING

```
1 print("Rows:", df.shape[0])
2 print("Columns:", df.shape[1])
3
4 df.info()
```

Rows: 7043
Columns: 21
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
Column Non-Null Count Dtype
--- ---
0 customerID 7043 non-null object
1 gender 7043 non-null object
2 SeniorCitizen 7043 non-null int64
3 Partner 7043 non-null object
4 Dependents 7043 non-null object
5 tenure 7043 non-null int64
6 PhoneService 7043 non-null object
7 MultipleLines 7043 non-null object
8 InternetService 7043 non-null object
9 OnlineSecurity 7043 non-null object
10 OnlineBackup 7043 non-null object
11 DeviceProtection 7043 non-null object
12 TechSupport 7043 non-null object
13 StreamingTV 7043 non-null object
14 StreamingMovies 7043 non-null object
15 Contract 7043 non-null object
16 PaperlessBilling 7043 non-null object
17 PaymentMethod 7043 non-null object
18 MonthlyCharges 7043 non-null float64
19 TotalCharges 7043 non-null object
20 Churn 7043 non-null object
dtypes: float64(1), int64(2), object(18)
memory usage: 1.1+ MB

CHECKING MISSING VALUES

```
1 df.isnull().sum()
```

	0
customerID	0
gender	0
SeniorCitizen	0
Partner	0
Dependents	0
tenure	0
PhoneService	0
MultipleLines	0
InternetService	0
OnlineSecurity	0
OnlineBackup	0
DeviceProtection	0
TechSupport	0
StreamingTV	0
StreamingMovies	0
Contract	0
PaperlessBilling	0
PaymentMethod	0
MonthlyCharges	0
TotalCharges	0
Churn	0

dtype: int64

Clean TotalCharges

```
1 df['TotalCharges'] = pd.to_numeric(df['TotalCharges'], errors='coerce')
2
3 df['TotalCharges'].isnull().sum()
```

np.int64(11)

Remove Missing Records

```
1 df = df.dropna()
2
3 df.shape
```

(7032, 21)

Churn Distribution

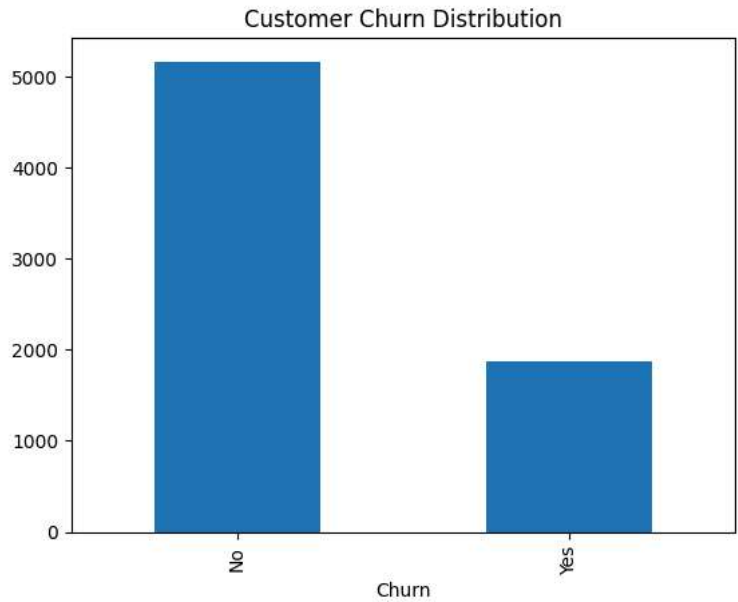
```
1 df['Churn'].value_counts()
```

	count
Churn	
No	5163
Yes	1869

dtype: int64

Plot

```
1 import matplotlib.pyplot as plt
2
3 df['Churn'].value_counts().plot(kind='bar')
4
5 plt.title('Customer Churn Distribution')
6 plt.show()
```



SQL Analysis

```
1 import sqlite3
2
3 conn = sqlite3.connect('telecom.db')
4
5 df.to_sql('customers',
6         conn,
7         if_exists='replace',
8         index=False)
```

7032

Query 1: Churn by Contract Type

```
1 query = """
2 SELECT Contract,
3 COUNT(*) AS Total_Customers
4 FROM customers
5 GROUP BY Contract
6 """
7
8 pd.read_sql(query, conn)
```

	Contract	Total_Customers	
0	Month-to-month	3875	
1	One year	1472	
2	Two year	1685	

Query 2: Average Monthly Charges

```
1 query = """
2 SELECT
3 AVG(MonthlyCharges) AS Avg_Monthly_Charge
4 FROM customers
```

```
5 ""
6
```

	Avg_Monthly_Charge
0	64.798208

Query 3: Churn Count

```
1 query = ""
2 SELECT
3 Churn,
4 COUNT(*) AS Customers
5 FROM customers
6 GROUP BY Churn
7 ""
8
9 pd.read_sql(query, conn)
```

	Churn	Customers
0	No	5163
1	Yes	1869

Encode target:

```
1 df['Churn'] = df['Churn'].map({
2     'No':0,
3     'Yes':1
4 })
```

```
1 df_encoded = pd.get_dummies(df,
2                               drop_first=True)
```

Split Data

```
1 from sklearn.model_selection import train_test_split
2
3 X = df_encoded.drop('Churn',
4                     axis=1)
5
6 y = df_encoded['Churn']
7
8 X_train, X_test, y_train, y_test = train_test_split(
9     X,
10    y,
11    test_size=0.2,
12    random_state=42
13 )
```

Train Model

```
1 from sklearn.ensemble import RandomForestClassifier
2
3 model = RandomForestClassifier()
4
5 model.fit(X_train, y_train)
```

▼ RandomForestClassifier ⓘ ?

RandomForestClassifier()

Evaluate

```
1 from sklearn.metrics import accuracy_score
2
```

```
3 predictions = model.predict(X_test)
4
5 accuracy_score(y_test,
6                 predictions)
```

0.7924662402274343

Exporting Clean Data

```
1 df.to_csv(
2     'cleaned_telco_data.csv',
3     index=False
4 )
```

DOWNLOADING CLEAN DATA

```
1 files.download(
2     'cleaned_telco_data.csv'
3 )
```